ANOMALY REPORT

Objective

The objective of anomaly detection is to automatically identify unusual or abnormal behavior in telecom network KPIs. Early identification of such anomalies helps network operators detect congestion, poor signal conditions, excessive packet loss, and other performance degradation before they significantly affect users.

KPIs Analyzed

The anomaly detection system considers the following network parameters:

KPI	**Significance**
RSRP	Indicates radio signal strength
SINR	Measures signal quality and interference
Latency	Represents network communication delay
Throughput	Measures network data transfer performance
Packet Loss	Indicates percentage of lost packets
Connected Users	Represents current network load

Isolation Forest Anomaly Detection

An Isolation Forest model was used as the first anomaly detection approach. Before model training, numerical KPI features were normalized to ensure comparable scales.

Isolation Forest identifies records that are significantly different from normal network observations. Anomalous records may contain conditions such as unusually high latency, low SINR, weak RSRP, high packet loss, or unexpected throughput degradation.

The model provides a computationally efficient, unsupervised approach because manually labeled anomaly data is not required.

LSTM-Based Anomaly Detection

An LSTM neural network was also implemented to detect time-series anomalies. Unlike Isolation Forest, LSTM considers the relationship between the consecutive KPI observations.

KPI records were converted into sequences using a sliding window of 20-time steps. The LSTM learned historical network behavior and predicted expected KPI values.

The prediction error was calculated using the Mean Squared Error (MSE).

Records having usually high prediction errors were classified as anomalies.

Threshold Tuning

Different prediction-error thresholds were evaluated.

Threshold	**Detection Behavior**	**False Positives**	**Missed Anomalies**
Low-90th percentile	Detects more anomalies	High	Low
Medium-95th percentile	Balanced detection	Medium	Medium
High-99th percentile	Detects severe anomalies	Low	High

The 95th percentile threshold was selected as a balanced approach between sensitivity and false alarms.

Anomaly Analysis

The detected anomalies mainly represent abnormal combinations of telecom KPIs.

Examples:

- Low RSRP – Possible poor radio coverage
- Low SINR – Possible radio interference
- High Latency – Possible network core/congestion
- High Packet Loss – Possible transport or backhaul problem
- Low Throughput – Possible heavy traffic or capacity limitation
- High Connected Users – Possible cell overload

The detected records were exported to CSV files so that they could be used by the Root Cause Analysis module and displayed on the final Streamlit dashboard.

Isolation Forest vs LSTM

Parameter	**Isolation Forest**	**LSTM**
Technique	Machine Learning	Deep Learning
Data Type	Individual KPI records	Sequential KPI data
Temporal Pattern Learning	No	Yes
Labels Required	No	No
Training Complexity	Low	Higher
Point Anomaly Detection	Good	Good
Sequential Anomaly Detection	Limited	Strong

Telecom Time-Series Suitability	Good	Very good

Result

Both models successfully provide mechanisms for identifying abnormal telecom network behavior. Isolation Forest is effective for quickly detecting unusual individual KPI records, whereas LSTM provides additional capability by learning temporal network patterns.

The detected anomalies can subsequently be passed to the Root Cause Analysis (RCA) module to determine probable network problems and recommend corrective actions.

Conclusion

The anomaly detection module provides an automated approach for identifying potential telecom network degradation. Combining Isolation Forest and LSTM allows the system to detect both individual KPI abnormalities and time-dependent anomalous behavior.

This supports proactive network monitoring by enabling operators to investigate problems before they lead to major QoS/QoE degradation.